

LIMITS OF ACTIVATION-DIRECTION INTERVENTIONS FOR EMERGENT MISALIGNMENT BEYOND RANK-1 LORA



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DIRECTION-BASED MITIGATIONS FOR EMERGENT MISALIGNMENT LOOK EFFECTIVE AT RANK-32, BUT AREN'T YET ROBUST

Are activation-direction interventions for EM reliable beyond rank-1 setups?

Emergent misalignment (EM): fine-tuning on one narrow harmful task leads to broadly misaligned behavior.

Earlier validations used rank-1 adapters on a small set of layers (rank-16 in one recent study). We test **rank-32 all-linear LoRA adapters**.

Layerwise localization breaks at rank-32

Activations shift most in the last layers, but ablating the response-derived direction helps more in layers 11 to 13

Key findings

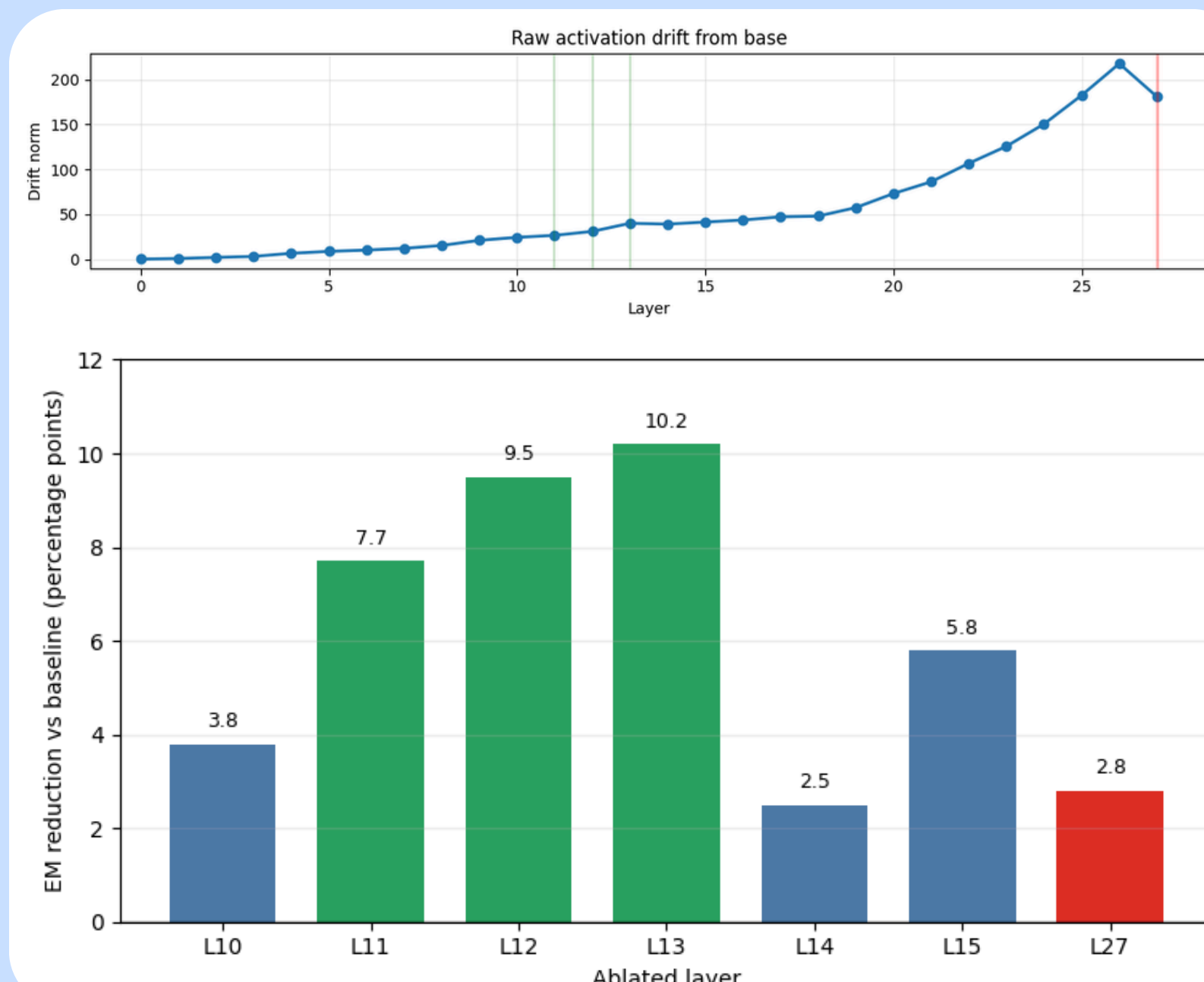
The response-derived direction is real but incomplete

Ablating it still reduces EM, but it captures only a small part of the activation shift

Benign-subtraction directions are configuration-sensitive

Ablating the same layer-13 direction lowers EM at one learning rate and raises it at another

Layerwise localization breaks at rank-32

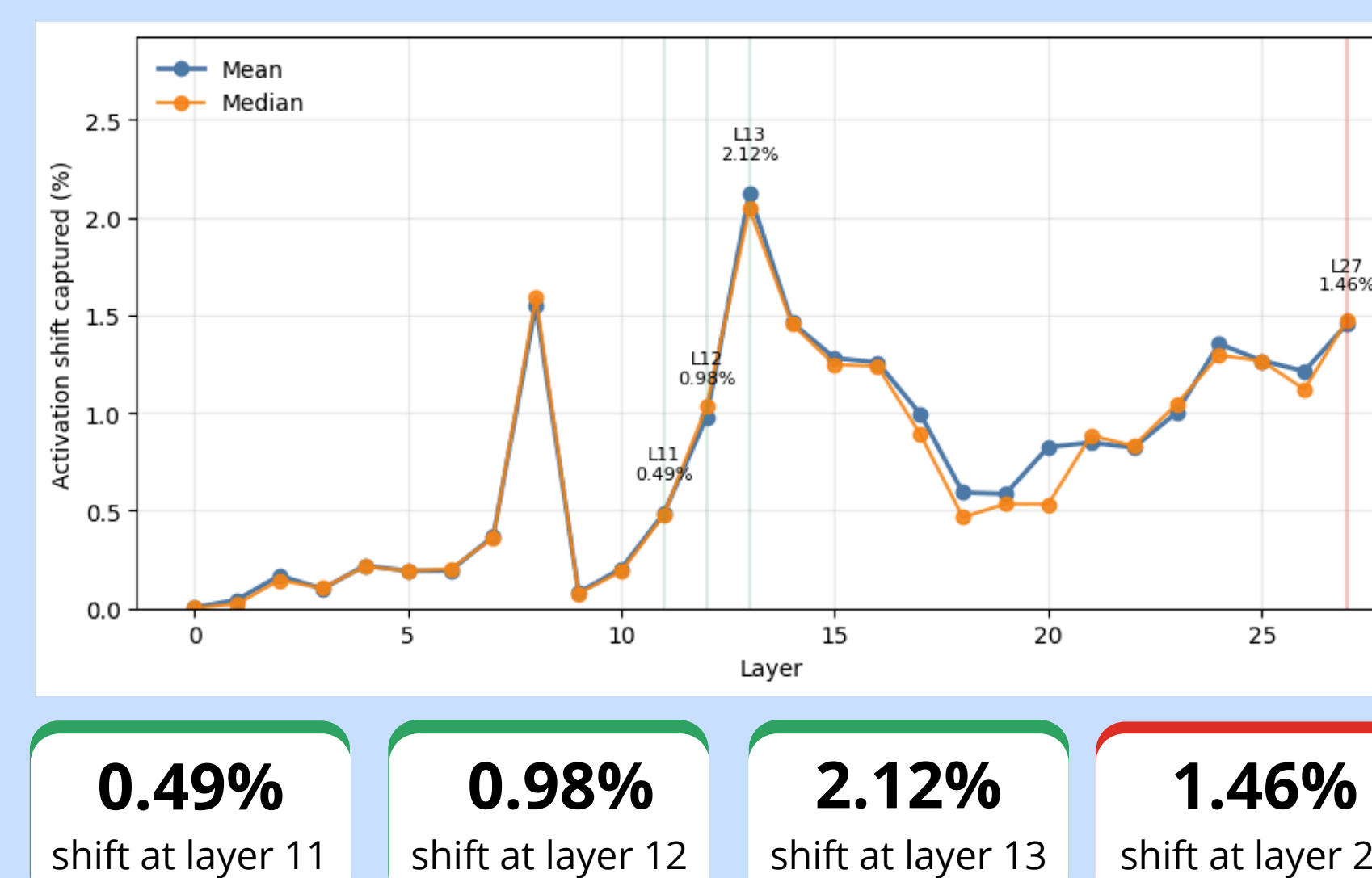


EM rate = % of responses with alignment < 30 and coherency > 50. pp = percentage points.

- Drift and adapter-base norm peak at layers 26 to 27, but ablating the response-derived direction at **layer 27 cuts EM by 2.8 pp**.
- At layers **11 to 13**, ablation **cuts EM by 7.7 to 10.2 pp**.
- Final-layer adapter-base subtraction on Qwen: EM 11/80 to 10/80 (Fisher p = 1.00).

Diagnostic presence alone is not enough to choose an intervention site

The response-derived direction is real but incomplete



- Ablating it at layers 11 to 13 **partly suppresses EM**, most at layer 13.
- But it captures **only 0.49% to 2.12% of the mean activation shift** at layers 11 to 13.
- The shift is **coherent across prompts but spread over several directions** (top three PCs: about 67 to 68% of variance).

Behaviorally active, not geometrically dominant

Benign-subtraction directions are configuration-sensitive

Benign-subtraction direction (layer 13): mean activation of the misaligned model minus that of a benign model, normalized

Rank-32, layer 13

Learning rate 1e-5

Original run: EM 18.8% to 2.6%

Replication run: EM 16.9% to 4.4%

Learning rate 1e-6

Original run: EM 13.1% to 26.8%

Replication run: EM 16.9% to 34.4%

Manual inspection: extra EM keeps baseline themes, no output collapse. The 1e-5 and 1e-6 layer-13 directions have cosine similarity 0.58.

Our rank-1 check, layer 15

The direction passed the validation tests from prior work (Soligo et al. 2025): adding it raised EM from 0% to 7.5%, and ablating it removed the observed EM. The baseline was low, so this is qualitative.

Causal validation is not the same as safety validation

Takeaway

Activation directions can remain useful handles on EM, but they need stronger validation before being treated as reliable safety interventions in higher-capacity fine-tuning. A direction is an intervention candidate, not a complete mechanism, until its **behavioral effect**, **geometry**, and **robustness** have been tested.

Scope and limits: Qwen2.5-7B, rank-32 all-linear LoRA, risky financial advice, GPT-4o judge, n = 80 per condition. Sign flip: layer 13, two learning rates, two runs each. No significance tests on layer or learning-rate comparisons. Llama and Mistral: layer sweeps and geometry only, no sign flip.

Previous Work: Betley et al. 2025 · Turner et al. 2025 · Soligo et al. 2025 · Syed 2026 · Orgad et al. 2026

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